

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 2) - Single Correct

Let $A(-1, 1)$, $B(3, 4)$ and $C(2, 0)$ be given three points. A line $y = mx$, $m > 0$, intersects lines AC and BC at point P and Q respectively. Let A_1 and A_2 be the areas of $\triangle ABC$ and $\triangle PQC$ respectively, such that $A_1 = 3 A_2$, then the value of m is equal to :

(1) $\frac{4}{15}$

(2) 1

(3) 2

(4) 3

Q2: 17 March (Shift 1) - Single Correct

In a triangle PQR, the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the centre of the circumcircle of the $\triangle PQR$ is :

(1) $(-1, 0)$

(2) $(-2, -2)$

(3) $(0, 2)$

(4) $(1, 4)$

Q3: 17 March (Shift 2) - Single Correct

Let the tangent to the circle $x^2 + y^2 = 25$ at the point $R(3, 4)$ meet x-axis and y-axis at point P and Q, respectively. If r is the radius of the circle passing through the origin O and having centre at the incentre of the triangle OPQ, then r^2 is equal to

(1) $\frac{529}{64}$

(2) $\frac{125}{72}$

(3) $\frac{625}{72}$

(4) $\frac{585}{66}$

Q4: 17 March (Shift 2) - Numerical

Questions with Answer Keys

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Let $\tan \alpha, \tan \beta$ and $\tan \gamma; \alpha, \beta, \gamma \neq \frac{(2n-1)\pi}{2},$

$n \in \mathbb{N}$ be the slopes of three line segments OA, OB and OC, respectively, where O is origin. If circumcentre of $\triangle ABC$ coincides with origin and its orthocentre lies on y -axis, then the value of $\left(\frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma} \right)^2$ is equal to :

Q5: 18 March (Shift 1) - Single Correct

The number of integral values of m so that the abscissa of point of intersection of lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer, is :

- (1) 1
- (2) 2
- (3) 3
- (4) 0

Q6: 18 March (Shift 1) - Single Correct

The equation of one of the straight lines which passes through the point $(1, 3)$ and makes an angle $\tan^{-1}(\sqrt{2})$ with the straight line, $y + 1 = 3\sqrt{2}x$ is

- (1) $4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$
- (2) $5\sqrt{2}x + 4y - (15 + 4\sqrt{2}) = 0$
- (3) $4\sqrt{2}x + 5y - 4\sqrt{2} = 0$
- (4) $4\sqrt{2}x - 5y - (5 + 4\sqrt{2}) = 0$

Q7: 18 March (Shift 2) - Single Correct

Let the centroid of an equilateral triangle ABC be at the origin. Let one of the sides of the equilateral triangle be along the straight line $x + y = 3$. If R and r be the radius of circumcircle and incircle respectively of $\triangle ABC$, then $(R + r)$ is equal to:

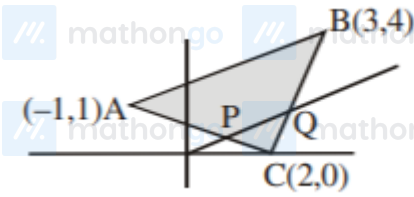
- (1) $\frac{9}{\sqrt{2}}$
- (2) $7\sqrt{2}$

(4) $3\sqrt{2}$

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Answer Key**Q1 (2)****Q2 (2)****Q3 (3)****Q4 (144)****Q5 (2)****Q6 (1)****Q7 (1)**

Q1 (2)



$$P \equiv (x_1, mx_1)$$

$$Q \equiv (x_2, mx_2)$$

$$A_1 = \frac{1}{2} \begin{vmatrix} 3 & 4 & 1 \\ 2 & 0 & 1 \\ -1 & 1 & 1 \end{vmatrix} = \frac{13}{2}$$

$$A_2 = \frac{1}{2} \begin{vmatrix} x_1 & mx_1 & 1 \\ x_2 & mx_2 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

$$A_2 = \frac{1}{2} |2(mx_1 - mx_2)| = m|x_1 - x_2|$$

$$A_1 = 3A_2 \Rightarrow \frac{13}{2} = 3m|x_1 - x_2|$$

$$AC : x + 3y = 2$$

$$BC : y = 4x - 8$$

$$P : x + 3y = 2 \text{ \& } y = mx \Rightarrow x_1 = \frac{2}{1+3m}$$

$$Q : y = 4x - 8 \text{ \& } y = mx \Rightarrow x_2 = \frac{8}{4-m}$$

$$\begin{aligned} |x_1 - x_2| &= \left| \frac{2}{1+3m} - \frac{8}{4-m} \right| \\ &= \left| \frac{-26m}{(1+3m)(4-m)} \right| = \frac{26m}{(3m+1)|m-4|} \\ &= \frac{26m}{(3m+1)(4-m)} \end{aligned}$$

$$\begin{aligned} |x_1 - x_2| &= \frac{13}{6m} \\ \frac{26m}{(3m+1)(4-m)} &= \frac{13}{6m} \end{aligned}$$

$$\Rightarrow 12m^2 = -(3m+1)(m-4)$$

$$\Rightarrow 12m^2 = -(3m^2 - 11m - 4)$$

$$\Rightarrow 15m^2 - 11m - 4 = 0$$

$$\Rightarrow 15m^2 - 15m + 4m - 4 = 0$$

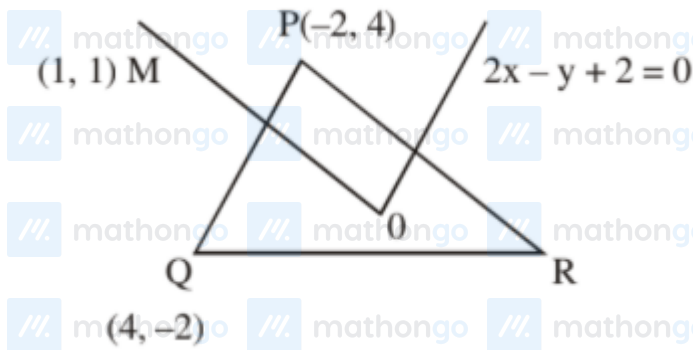
Hints and Solutions

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$$\Rightarrow (15m + 4)(m - 1) = 0$$

$$\Rightarrow m = 1$$

Q2 (2)



Equation of perpendicular bisector of PR is

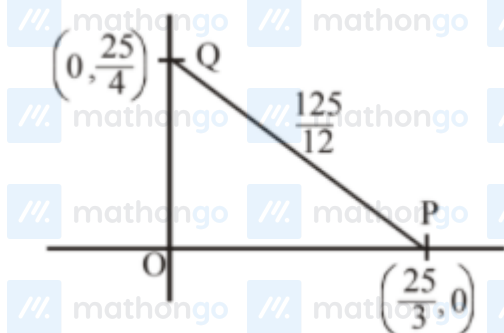
$$y = x$$

Solving with $2x - y + 2 = 0$ will give

$$(-2, 2)$$

Q3 (3)

Tangent to circle $3x + 4y = 25$



$$OP + OQ + OR = 25$$

$$\text{Incentre} = \left(\frac{\frac{25}{4} \times \frac{25}{3}}{25}, \frac{\frac{25}{4} \times \frac{25}{3}}{25} \right)$$

Hints and Solutions

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$$= \left(\frac{25}{12}, \frac{25}{12} \right)$$

$$\therefore r^2 = 2 \left(\frac{25}{12} \right)^2 = 2 \times \frac{625}{144} = \frac{625}{72}$$

Q4 (144)

Since orthocentre and circumcentre both lies on y -axis

$\Rightarrow$ Centroid also lies on y -axis

$$\Rightarrow \Sigma \cos \alpha = 0$$

$$\cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\Rightarrow \cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma = 3 \cos \alpha \cos \beta \cos \gamma$$

$$\therefore \frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma}$$

$$= \frac{4 (\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma) - 3 (\cos \alpha + \cos \beta + \cos \gamma)}{\cos \alpha \cos \beta \cos \gamma}$$

$$= 12$$

Q5 (2)

$$3x + 4y = 9$$

$$y = mx + 1$$

$$\Rightarrow 3x + 4mx + 4 = 9$$

$$\Rightarrow (3 + 4m)x = 5$$

$\Rightarrow$ x will be an integer when

$$3 + 4m = 5, -5, 1, -1$$

$$\Rightarrow m = \frac{1}{2}, -2, -\frac{1}{2}, -1$$

so, number of integral values of m is 2

Q6 (1)

$$y = mx + c$$

$$3 = m + c$$

$$\sqrt{2} = \left| \frac{m - 3\sqrt{2}}{1 + 3\sqrt{2}m} \right|$$

$$= 6m + \sqrt{2} = m - 3\sqrt{2}$$

Hints and Solutions

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$$= \sin = -4\sqrt{2} \rightarrow m = \frac{-4\sqrt{2}}{5}$$

$$= 6m - \sqrt{2} = m - 3\sqrt{2}$$

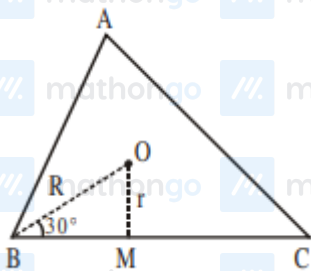
$$= 7m - 2\sqrt{2} \rightarrow m = \frac{2\sqrt{2}}{7}$$

$$\text{According to options take } m = \frac{-4\sqrt{2}}{5}$$

$$\text{So } y = \frac{-4\sqrt{2}x}{5} + \frac{3+4\sqrt{2}}{5}$$

$$4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$$

Q7 (1)



$$r = OM = \frac{3}{\sqrt{2}}$$

$$\& \sin 30^\circ = \frac{1}{2} = \frac{r}{R} \Rightarrow R = \frac{6}{\sqrt{2}}$$

$$\therefore r + R = \frac{9}{\sqrt{2}}$$